Solar ESS Co. Ltd.
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ELECTRICIAN JOB SHEET · PROJECT BASED · QUOTE 2 OPTIONS

Battery backup for the water pump and lodge loads: manual ATS (Option A) or MTS for the pump (Option B)

Harmony@Huailan Ecolodge, Chiang Mai · Job no. JOB-HHL-PUMP-01 · Issued by Solar ESS Working Group · 2 Nov 2026

How it is wired today

- PEA grid → meter → **inverter GRID port**, and from the same point to the **MDB → all loads** (water pump, water heater and the other circuits).
- The inverter **BACKUP (LOAD) port** carries only 30–130 W. In a blackout the pump and all loads go off.
- The inverter can supply **10 kW total / 5 kW per phase** on backup (2× surge for 10 s), but the water heater alone draws 11–12 kW.

Option A: ATS at the MDB, kept in manual – back up the whole MDB

- PEA grid (after the inverter GRID connection) → **ATS source A**.
- Inverter BACKUP port → new MCB 4P 32 A (inverter maker's rating) → **ATS source B**.
- ATS output → MDB main breaker → all loads, including the pump.
- **Keep the ATS in MANUAL mode**: the total load can be higher than the inverter can supply. A contactor (K1) in the water-heater circuit opens automatically whenever the ATS is on B, so the heater can never run on the battery. Switch off other big loads before turning to B.

Option B: 32 A MTS (manual I–0–II) for the water pump only

- Inverter BACKUP port → new MCB 2P 32 A → **MTS position II**.
- Existing pump MCB in the MDB (grid) → **MTS position I**.
- MTS output → RCBO 30 mA, rated the same as the existing pump MCB (typically 16–20 A) → water pump.
- In a blackout, turn the MTS to II to run the pump from the battery; when the grid is back, turn it back to I. Reference switch: Schneider GoPact MTSE open transition (I–0–II) or equal.

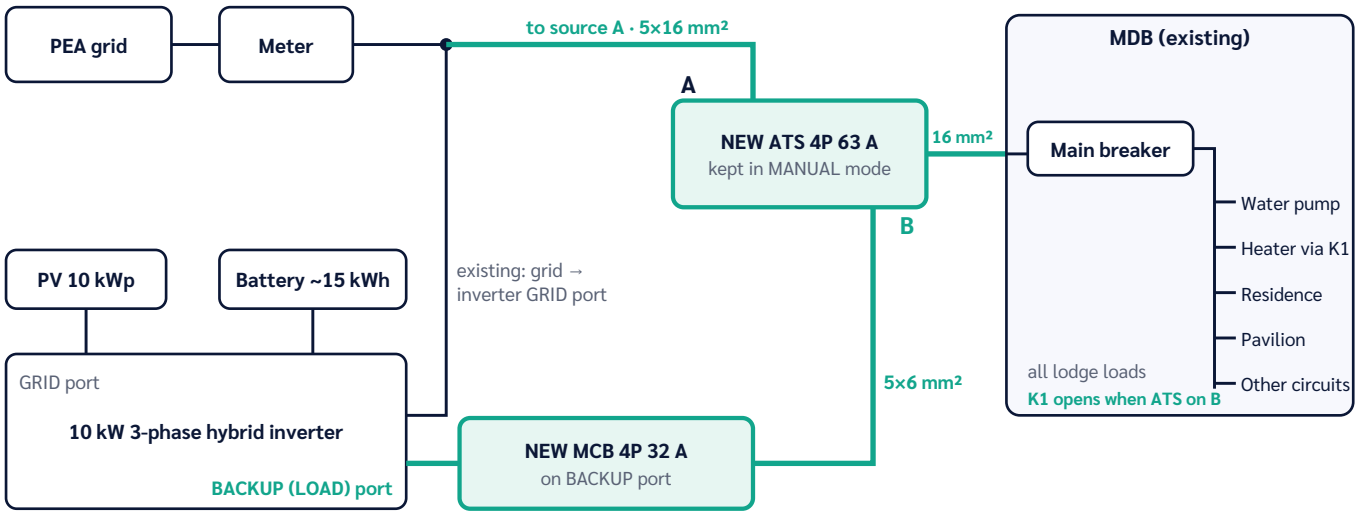
Rules that must never be broken: never connect the BACKUP port in parallel with the grid. Every switch must be break-before-make with a mechanical interlock and must switch the neutral too. Never connect the inverter GRID port to the output side of the ATS/MTS.

Existing system on site

Item	Details
Solar panels	10 kWp
Inverter	10 kW three-phase low-voltage hybrid · BACKUP port max 10 kW total / 5 kW per phase
Battery	Lithium 48 V 300 Ah (~15 kWh) · set to discharge down to 25%
MDB	Steel MDB with volt/ammeters; feeders: Residence, Pavilion, Guest House, Outside Toilet and spares
Big loads	Two-phase water heater ~11–12 kW (a few minutes a day) and the water pump (confirm rating on site)

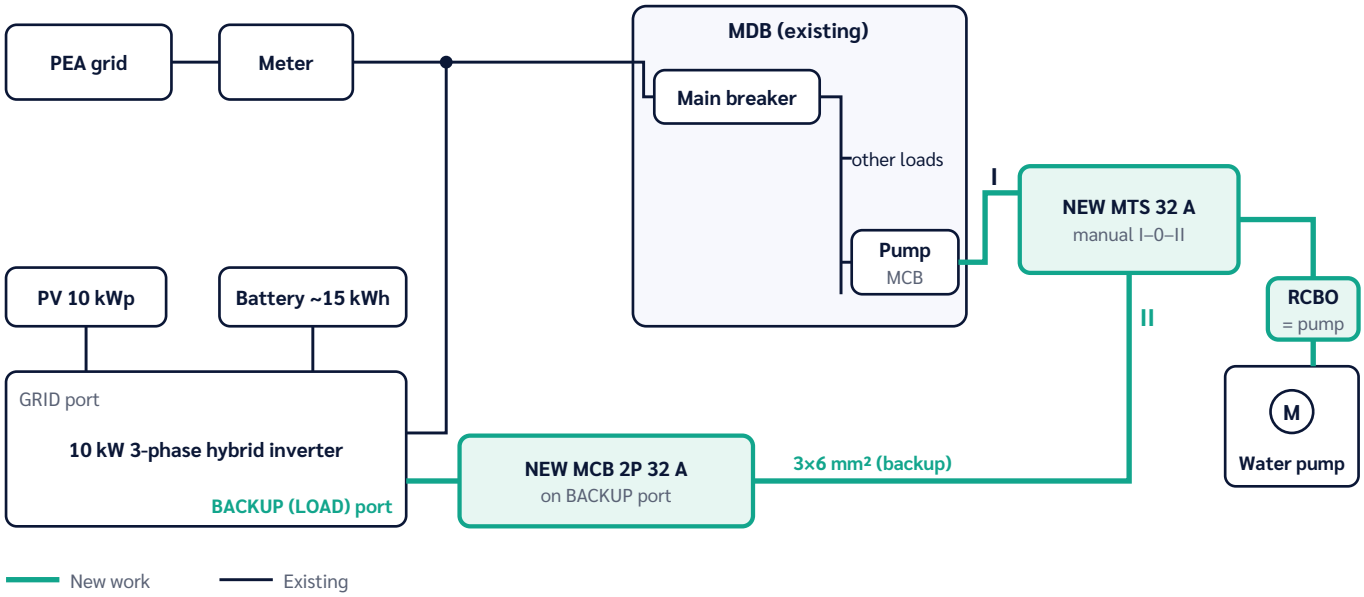
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Single-line diagram – Option A



Source A and the ATS output carry the whole lodge from the grid (peaks about 27–29 A per phase with the heater on), so they are sized to the grid side: ATS 4P 63 A (at least the upstream grid breaker) and THW 16 mm². Source B carries only the inverter (max 22.7 A per phase), so 4P 32 A and THW 6 mm² as the inverter maker recommends. Phase sequence on sources A and B must match.

Single-line diagram – Option B



Shown for a single-phase pump. For a three-phase pump use all 4 poles of the MTS, a 4P MCB and 5-core cable.

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Scope of work – Option A

1. **Survey:** MDB main breaker and main cable size, where the inverter GRID port is connected, inverter CT position, ATS location and cable runs. Send to Solar ESS before buying materials.
2. **Isolate (LOTO):** grid main, inverter GRID and BACKUP AC breakers, inverter off, battery breaker and PV DC isolators. Prove dead.
3. **ATS:** install a 4P 63 A ATS (not below the upstream grid breaker) with an auxiliary contact, in an IP54 enclosure beside the MDB. **Set it to MANUAL** and disable automatic transfer.
4. **Source A:** move the MDB incoming main cable (downstream of the inverter GRID connection) to ATS source A with THW 5×16 mm² (or the existing main size if larger). The inverter GRID port must stay connected directly to the grid as today.
5. **Source B:** install a new MCB 4P 32 A on the BACKUP port and run THW 5×6 mm² (L1 L2 L3 N PE) in 1" IMC/EMT to ATS source B.
6. **ATS output:** THW 5×16 mm² (or existing main size) to the MDB main breaker.
7. **Heater lockout (K1):** install contactor K1 (40 A AC-1, poles to suit the heater) in the water-heater circuit. Feed its coil through the ATS auxiliary contact "position A", so K1 opens whenever the ATS is on B.
8. **Inverter CTs:** must stay on the grid cable upstream of the ATS (source A side). Check CT direction afterwards.
9. **Neutral:** the BACKUP-port neutral passes through the ATS only; never join it directly to the MDB neutral bar. PE may be common.
10. **Operating label at the ATS:** "Blackout: 1) switch off big loads 2) turn ATS to B (heater locks out) · Grid back: turn ATS to A" and "Backup max 10 kW / 5 kW per phase".
11. **Test and hand over** per the test table, and show the lodge owner how to operate it.

Scope of work – Option B

1. **Survey:** pump nameplate (kW, V, 1- or 3-phase, starting method), starting current, pump circuit in the MDB and cable runs.
2. **Isolate (LOTO)** as in Option A.
3. **MTS:** install a 32 A MTS (I–0–II, open transition, e.g. Schneider GoPact MTSE or equal) in an IP54 enclosure at the MDB or near the pump.
4. **Position I:** move the outgoing cable of the existing pump MCB in the MDB to MTS position I.
5. **Position II:** new MCB 2P 32 A on the BACKUP port (least-loaded phase, L1 recommended); THW 3×6 mm² in ¾" conduit to MTS position II (10 mm² if over 30 m).
6. **Output:** MTS → RCBO 1P+N 30 mA rated the same as the existing pump MCB (typically 16–20 A, never above the pump cable rating) → water pump; earth the pump. The pump must keep its own motor overload protection.
7. **Label:** "Blackout: turn to II (battery) · Grid back: turn to I (grid)".
8. **Test and hand over** per the test table, and show the lodge owner how to operate it.

> **Danger:** the inverter BACKUP port stays live from the battery and the solar panels even when the grid is off. Always switch off the inverter, the battery and the DC isolators first, lock out and tag out (LOTO), and prove dead. Tell the lodge owner in advance: Option A needs the whole lodge switched off while the main cable is moved.

Materials (minimum specification)

Premium materials only. Give the brand and size of every item in your quote.

Option	Item	Minimum specification	Qty
A	ATS	4P 63 A (not below the upstream grid breaker), 400 VAC, mechanical interlock, break-before-make, manual mode, 1 auxiliary contact	1
A	Heater contactor K1	40 A AC-1, 230 V coil, poles to suit the heater circuit	1
A	ATS enclosure	Steel IP54 with DIN rail and terminals	1
A	MCB on BACKUP port	4P 32 A curve C 6 kA	1
A	Cable	THW 6 mm² (5 cores) BACKUP port → ATS; THW 16 mm² (5 cores) for source A and the ATS output; 1.5 mm² control wire for K1	As measured
A	Conduit	1" and 1¼" IMC/EMT, with fittings, saddles and pull boxes	As measured
B	MTS	32 A 415 V, I–0–II open transition, IEC 60947-6-1, AC-23A (ref. Schneider GoPact MTSE or equal); 4P, or 2 poles used for a 1-phase pump	1
B	MTS enclosure	IP54 with DIN rail and terminals	1
B	MCB on BACKUP port	2P 32 A curve C 6 kA	1
B	RCBO, pump side	1P+N 30 mA Type A, same rating as the existing pump MCB (typically 16–20 A)	1
B	Cable and conduit	THW 6 mm² (black, blue, green/yellow) in ¾" IMC/EMT; waterproof flexible connection at the pump	As measured
A/B	Consumables	Lugs, ferrules, cable markers, operating and warning labels, cable ties, anchors	1 set

Tests (record every reading)

Option	Test	Pass when
A/B	Insulation of every new cable at 500 VDC	≥ 1 MΩ
A/B	Polarity and earth continuity	Correct
A	Phase sequence of source A and source B is the same (essential for 3-phase motors/pumps)	Same
A	ATS on A: all loads on the grid, inverter normal, CTs read in the right direction	Normal
A	Heater lockout: with ATS on B, switch the heater on; K1 must stay open	Heater stays off
A	Simulated blackout: open the grid main, turn ATS to B; run pump and lights 15 min; record power per phase	≤ 5 kW/phase, no trip
A	Grid back: turn ATS to A; inverter resynchronises with the grid	Normal
B	MTS on I: pump runs on the grid as before	Pump normal
B	Simulated blackout: open the grid main, turn MTS to II; pump runs on battery 10 min, restart 3 times, record starting current	No trip
B	RCBO test button	Trips at once

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To confirm on site

- MDB main breaker, grid supply breaker and main cable size (the ATS and source-A cable must not be smaller than the upstream breaker allows).
- Water pump: kW, 1- or 3-phase, existing pump MCB and cable size, starting current. If the direct-on-line starting current is above about 40 A, tell Solar ESS: a soft starter is needed for battery operation.
- Inverter CT position and the existing terminal block (labels RM1, SM1, TM1, RKM, SKM, TKM, LLM, N): identify every conductor before touching it.
- Cable runs and the ATS/MTS mounting position.

How to quote

- Quote **Option A and Option B separately**.
- **Two prices for each option:** 1) Labour cost only; 2) Labour & materials (list every material by brand and size).
- Pay guide for labour: **Option A ฿6,000** (2 people, ~2 days) · **Option B ฿3,000** (2 people, ~1 day); materials and travel not included.
- Attach your electrician certificates and photos of similar work. Do not change any inverter settings; Solar ESS sets the inverter remotely.
- Send your quote with the form on www.solaress.org/job.html.

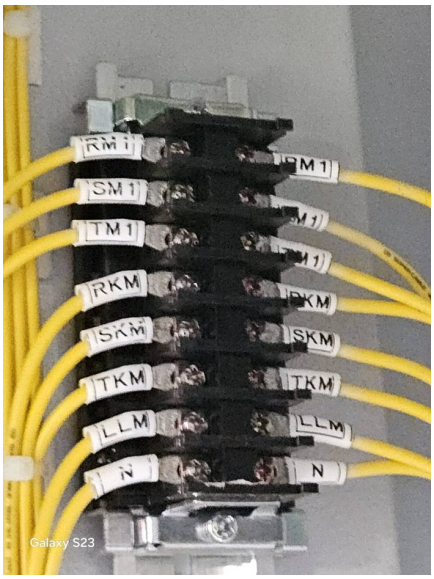
Site photos



Existing MDB (the ATS/MTS goes here)



Inverter room with AC and PV boxes



Existing terminal block (identify before work)

Qualifications and certificates of the Solar ESS team are furnished upon request or upon completion of contract.

Solar ESS Working Group